

Unit – I Programming Methodology

Problem:

A problem is a situation that is unsatisfactory and causes difficulties for people.

A problem means anything, matter, person, etc, that is difficult to deal with, solve, or overcome.

A problem is any thing or situation which is difficult to understand or deal with or handle it.

E.g1) A person who wants addition of two numbers 2) A person who wants to find out area of circle.

Problem solving steps:

Programmer cannot write instructions directly unless he knows how to solve any problem. To solve problem following steps are used:

- 1) Analysis of Program 2) Planning the program
- 3) Coding the program 4) Running or Debugging the program

1) Analysis of Program :

In this step the programmer must have to know which type of data is required to solve any problem.

e.g. To find addition of two numbers, two numbers are the data.

2) Planning the Program :

In this step which methods or operations are performed on to the data is decided to solve the given problem.

e.g. for addition of two numbers, addition operation is required.

3) Coding the Program :

After planning the program, the programmer can write the code/instructions of program by using programming languages.

4) Running or Debugging the program :

After coding the program, the programmer can run or debug (step by step execution of the program to get output are known as running), if the correct output is not get then programmer can check the errors and remove it are known as debugging.

Logic:

The steps /solutions required for solving the problems known as logic.

Types of logic:

Following are three different types of logic

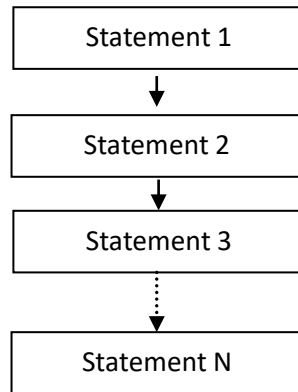
- 1) Sequence Logic 2) Selection(Decision making) Logic 3) Iteration Logic

Let's see all these logics briefly...

1) Sequence Logic:

In sequence logic all instructions or statements are executed in sequence manner as they appear. That is from top to bottom, one after another in sequence manner.

Working of sequence logic is shown in following fig:



e.g Following example uses the sequence logic

step 1: Start

step 2: read the value of a

step 3: read the value of b

step 4: calculate $c=a+b$;

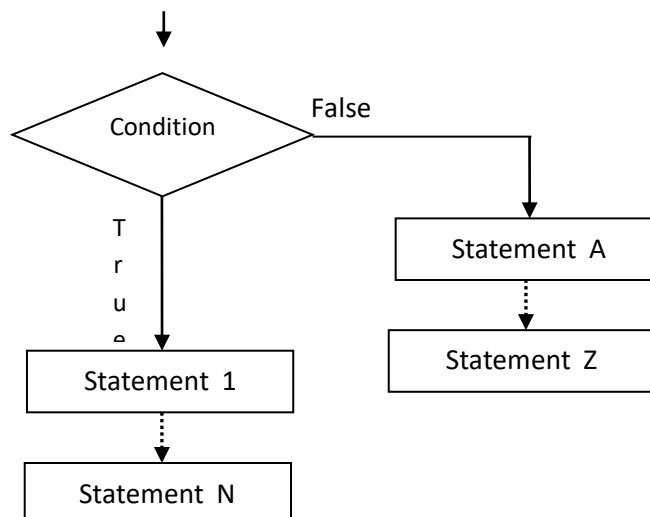
step 5: print c

step 6: Stop

1) Selection (Decision making) Logic:

Selection logic is also known as “Decision Logic” which is used to make certain decision. To solve problem, in some situation we have to check condition for that selection logic is used. This logic selects or chooses correct option or choice from number of choices. It uses If-Then, If-Then-Else or switch statements.

Working of selection logic is shown in following fig:



e.g. Following example uses the selection logic.

1) Write the algorithm to check entered number is positive or negative.

Step 1: Start

Step 2: Read N

Step 3: If ($N > 0$) Then

Display 'N' is Positive

Else

Display 'N' is Negative

Step 4: Stop

2) Iteration Logic (Looping):

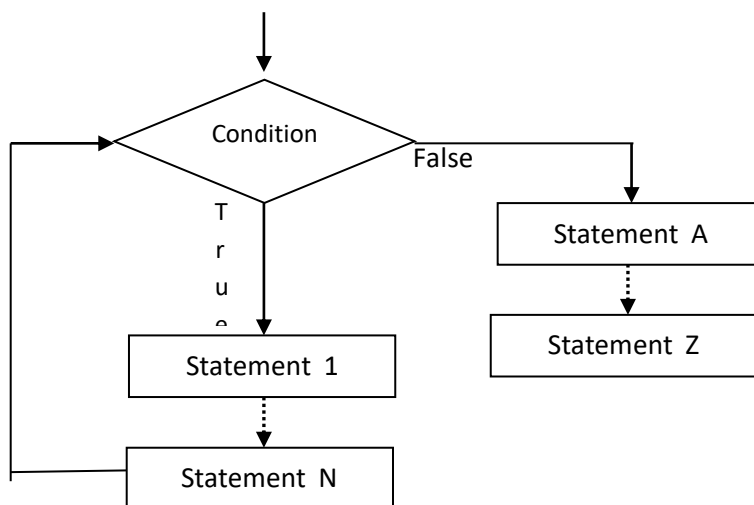
Iteration logic is also known as “Looping”. In some situation we have to repeat some statements again and again at that time Iteration logic i.e. looping is used.

Looping:

“It is the process to execute some statements repeatedly until the condition becomes false.”

The looping is continue as long as the condition is True and stops when condition becomes false. This logic uses Do-while, while, for, Repeat-until statements.

Working of Iteration logic is shown in following fig:



e.g. Following example uses the Iteration logic.

1) Write the Pseudo code to Print series like 1 2 3 4 5.

Pseudo Code:

```
Set      1 to N
While (N <= 5)
    Display N
    Add    1 to N
End While
Stop
```

- **Programming Planning Tools :**

“Programming planning tools are used to plan or build or grow the program logic of the programmer.”

There are three kinds of programming planning tools are used :

1) Algorithm 2) Flowchart 3) Pseudo code

- **1) Algorithm:**

“Algorithm is the step by step description of how to solve any problem”

In short algorithm is nothing but logic of program. It contains the sequence of instructions which would after execution gives desired output.

- **Characteristics (Features) of the Algorithm :**

- 1) Each and every instruction should be accurate and unmistakable.
- 2) One or more instructions should not be repeated.
- 3) The desired result must be obtained.
- 4) It is easy to understand, because representation of algorithm is in very simple English language.
- 5) It gives the language independent layout of the program.

e.g. 1) Algorithm for addition of two numbers.

Step 1 : Start

Step 2 : Input any two numbers say ‘A’ and ‘B’

Step 3 : Calculate,

$$C = A + B$$

Step 4 : Display ‘C’

Step 5 : Stop.

2) Algorithm to find area of rectangle.

Step 1 : Start

Step 2 : Input any two numbers say 'length' and 'breadth'

Step 3 : Calculate,

$$\text{area} = \text{length} * \text{breadth}$$

Step 4 : Display 'area'

Step 5 : Stop.

2) Flowchart:

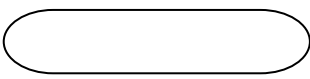
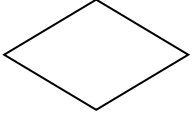
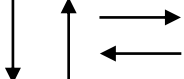
Flowchart is also programming planning tool which is used to build or grow the program logic of programmer.

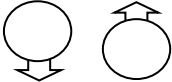
Definition:

“Flowchart is pictorial or symbolic or diagrammatical representation of algorithm”

That is we represent algorithm by using different symbols and every symbols having their own meaning.

The symbols and their meaning is as follows:

Symbol Name	Symbol	Meaning or Use
Terminal Symbol		This symbol is used to show <u>start and stop</u> step of algorithm.
Input-Output Symbol		This symbol is used to show <u>input and output</u> step of algorithm.
Process Symbol		If there are some processes such as addition, subtraction, multiplication, division, modulation, assignment (Initialization) etc. then it is shown by using process symbol.
Decision Making Symbol		If there is a situation to take decision or if we have to check particular condition then it is shown by using decision making symbol.
Flow Lines		The flow lines are used to represent the flow of problem.

Connector Symbol		This symbol is used to connect two flowcharts together.
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Advantages or Characteristics or Principles of Flowchart:

- 1) It is easy to understand, because it is symbolic representation of algorithm.
- 2) By using flowchart, the programmer gets flow of program.
- 3) It allows easy debugging of program.
- 4) By using flowchart, we can write program code very easily.
- 5) By using flowchart, programmer can concentrate on program logic.
- 6) It detects errors easily.

e.g. 1) Draw the flowchart and draw the flowchart for addition of two numbers.

Algorithm:

Step 1 : Start

Step 2 : Input any two numbers say 'a' and 'b'

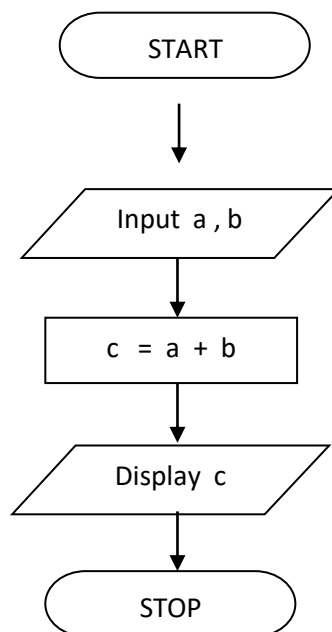
Step 3 : Calculate,

$$c = a + b$$

Step 4 : Display 'c'

Step 5 : Stop.

Flowchart:



3) Pseudo code:

Pseudo code is programming planning tool that helps programmer to plan or build or grow the program logic by writing instructions or code in an ordinary natural languages such as English, Chinese, Japanese, French etc.

The word Pseudo code consist from two words namely Pseudo and code.

Pseudo means Imitation or False and Code means Instructions or statements.

Characteristics or Principles of Pseudo code:

- 1) it is easy to understand, because it uses the natural languages to solve the problem
- 2) it does not have any syntax therefore programmer solve problem very easily.
- 3) it allows programmer to concentrate on program logic.
- 4) By using Pseudo Code the programmer can easily convert program logic into programming languages.

e.g. Following example uses the sequence logic.

- 1) Write the Pseudo code to find addition of two numbers

Pseudo Code:

```
Read  a and b
Calculate,      C = a + b
Display C
Stop
```

e.g. Following example uses the selection logic.

- 1) Write the Pseudo code to check entered number is positive or negative.

Pseudo Code:

```
Read  N
If (N > 0) Then
    Display 'N' is Positive
Else
    Display 'N' is Negative
End if
Stop
```

e.g. Following example uses the Iteration logic.

- 1) Write the Pseudo code to Print series like 1 2 3 4 5.

Pseudo Code:

```
Set      1 to N
While (N <= 5)
    Display N
    Add    1 to N
End While
```

Stop

Examples on algorithm for Home work:

- 1) Write an algorithm to find average of three numbers.
- 2) Write an algorithm to calculate Perimeter (Circumference) of Circle.
- 3) Write an algorithm to calculate area of Circle.
- 4) Write an algorithm to calculate Perimeter (Circumference) of Circle.
- 5) Write an algorithm to calculate area of triangle.
- 6) Write an algorithm to calculate area of square.
- 7) Write an algorithm to check entered number is positive or negative.
- 8) Write an algorithm to check entered number is Even or Odd.
- 9) Write an algorithm to find minimum number between two numbers.
- 10) Write an algorithm to find maximum number between two numbers.
- 11) Write an algorithm to find minimum number between three numbers.
- 12) Write an algorithm to find maximum number between three numbers.
- 13) Write an algorithm to find digit sum of entered number.
- 14) Write an algorithm to check entered number is Armstrong or not.
- 15) Write an algorithm to find reverse number of entered number.
- 16) Write an algorithm to check entered number is Palindrome or Not.
- 17) Write an algorithm to find factorial of entered number.

Examples on flowchart for Home work:

- 1) Draw the flowchart to find average of three numbers.
- 2) Draw the flowchart to calculate Perimeter (Circumference) of Circle.
- 3) Draw the flowchart to calculate area of Circle.
- 4) Draw the flowchart to calculate Perimeter (Circumference) of Circle.
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- 6) Draw the flowchart to calculate area of square.
- 7) Draw the flowchart to check entered number is positive or negative.
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- 9) Draw the flowchart to find minimum number between two numbers.
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- 15) Draw the flowchart to find reverse number of entered number.
- 16) Draw the flowchart to check entered number is Palindrome or Not.
- 17) Draw the flowchart to find factorial of entered number.

Assignment No. 1

- 1) What is Program? Explain different steps to solve the problem.
- 2) What is algorithm? What are the different characteristics or principles of algorithm? Explain algorithm with suitable example.
- 3) What is Flowchart? What are the advantages of flowchart over algorithm? Explain it with suitable example.
- 4) Define Flowchart. Explain different symbols used in flowchart.
- 5) List out flowcharting principles.
- 6) What is Programming Planning tool? List out different programming planning tools with their definition.
- 7) Explain Pseudo code in details.
- 14) What are the different types of Logics? Explain all logics in details.

